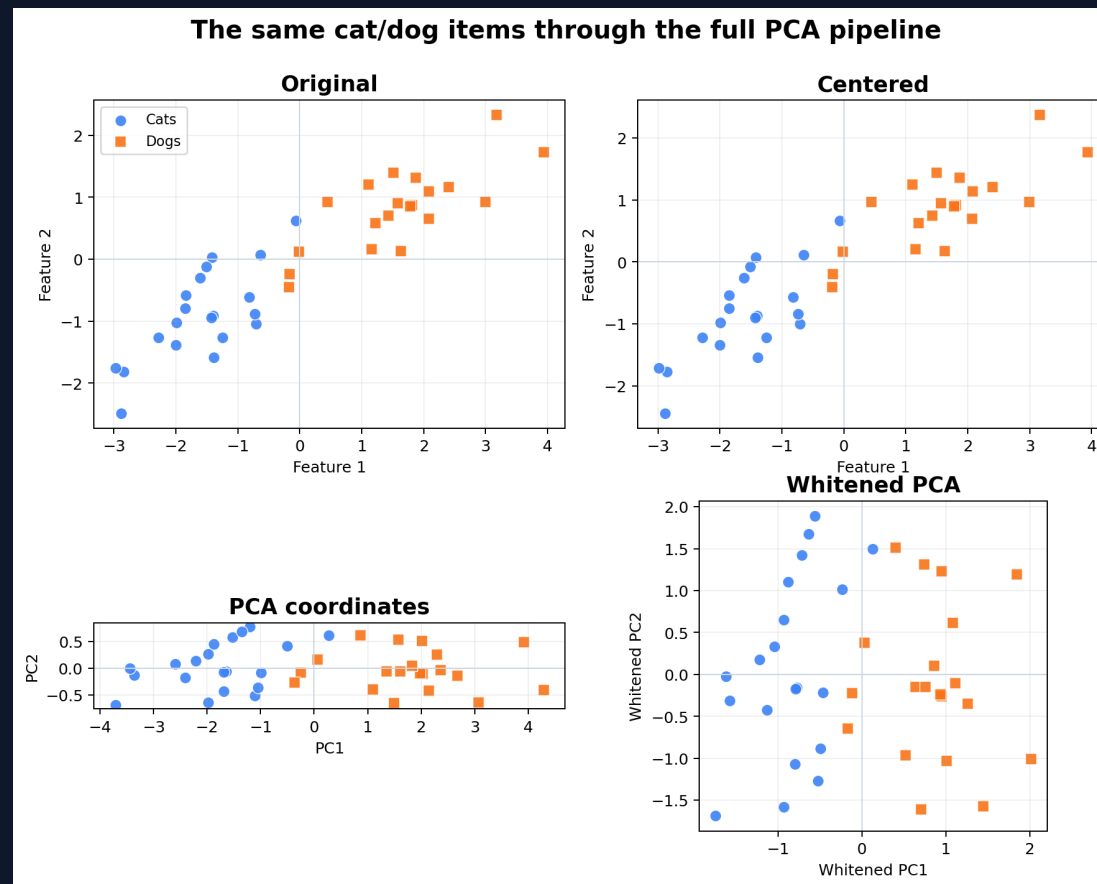


PCA, Eigenvectors, Eigenvalues & Whitening

Cat vs Dog - all graph states, step by step

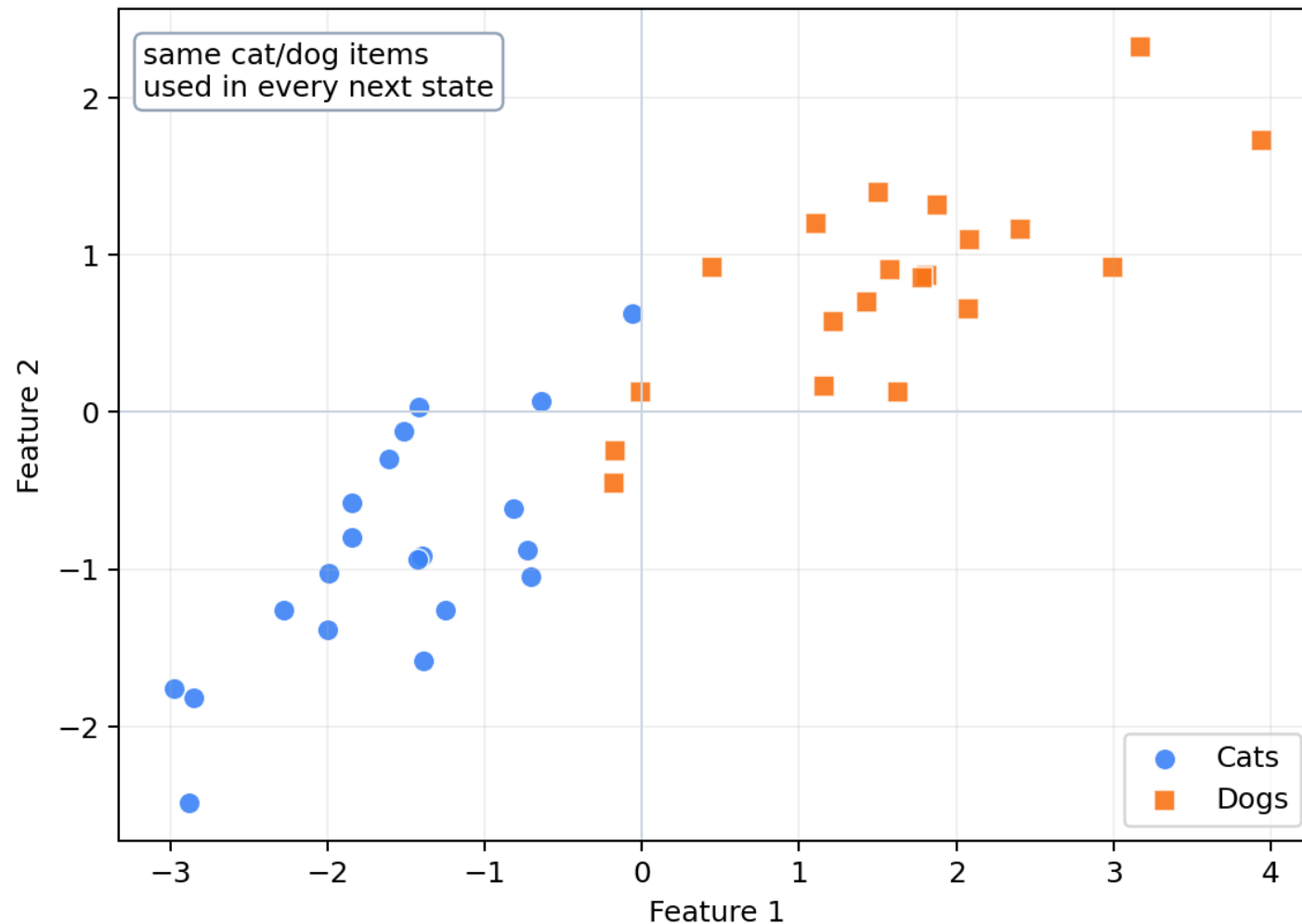


Illustrative 2D embeddings created to explain the geometry clearly; the same points are transformed on every page.

1. Original embedding space

Same points - transformed state by state

State 1 - Original 2D embedding space

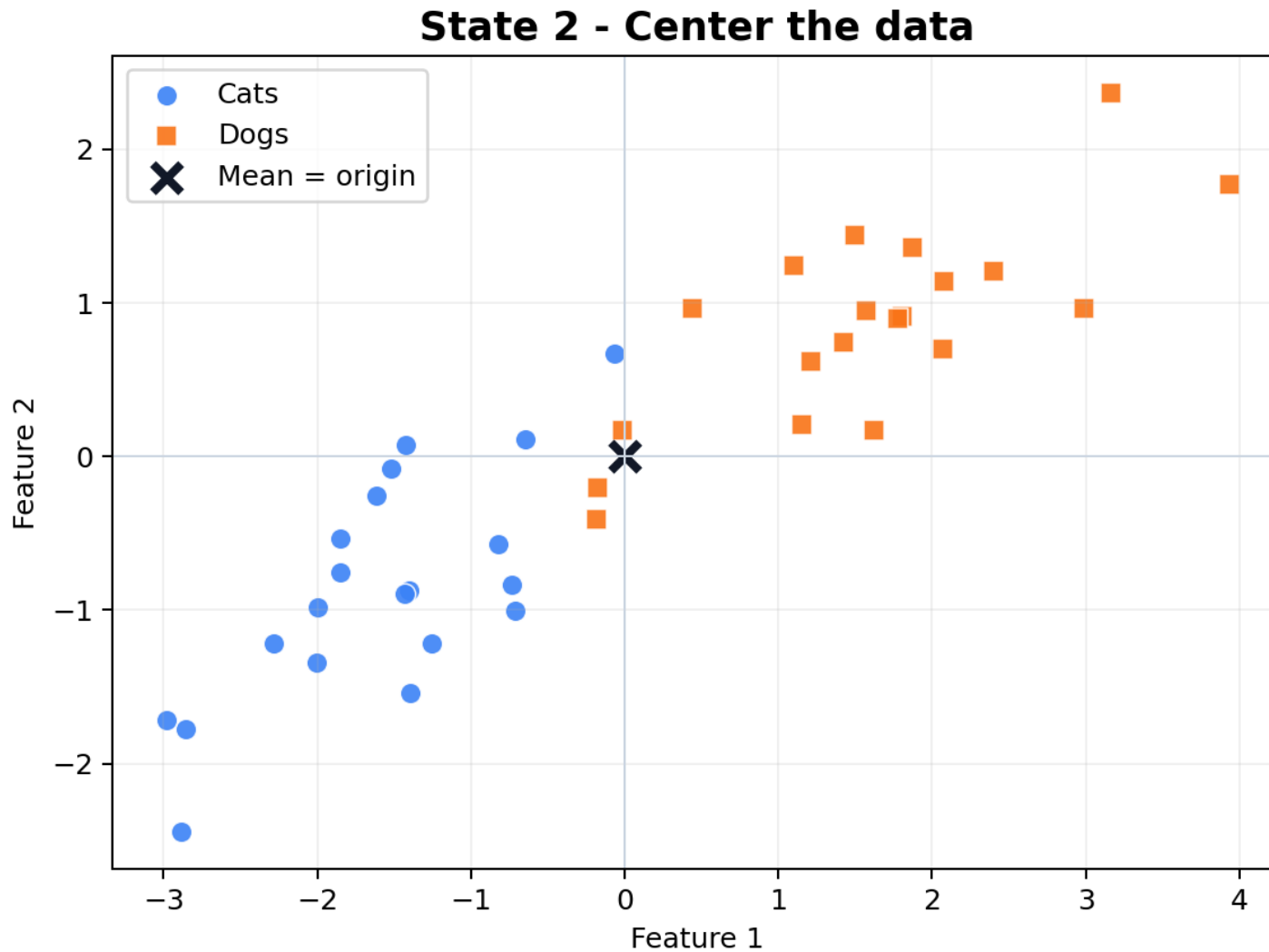


What this state means

- Each dot is one item embedding.
- Blue circles are cats; orange squares are dogs.
- The cloud is elongated: there is more spread in one diagonal direction.

2. Center the data

Same points - transformed state by state

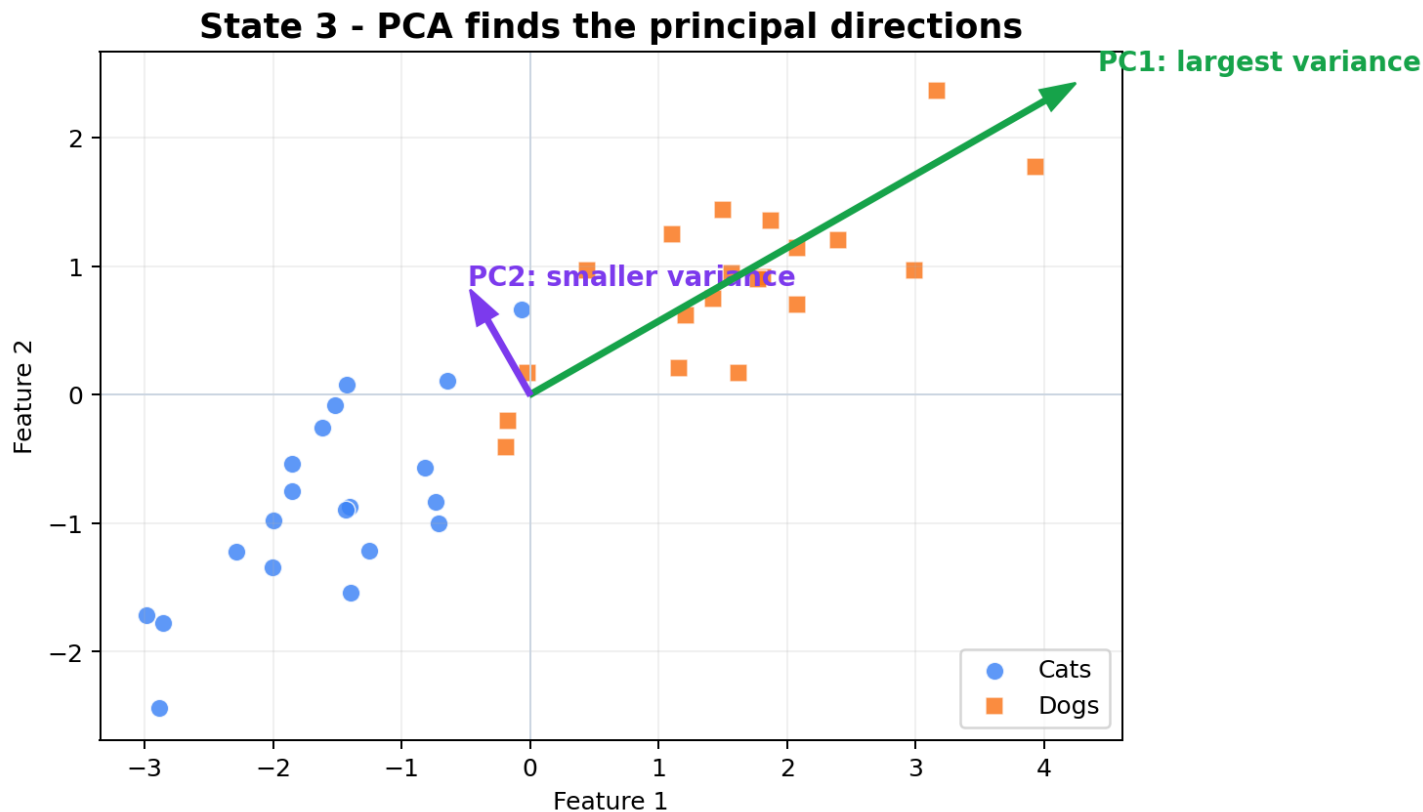


What this state means

- Subtract the mean vector from every point.
- The geometry and cat/dog relationships do not change.
- PCA is then computed around the origin.

3. Find eigenvectors / principal components

Same points - transformed state by state



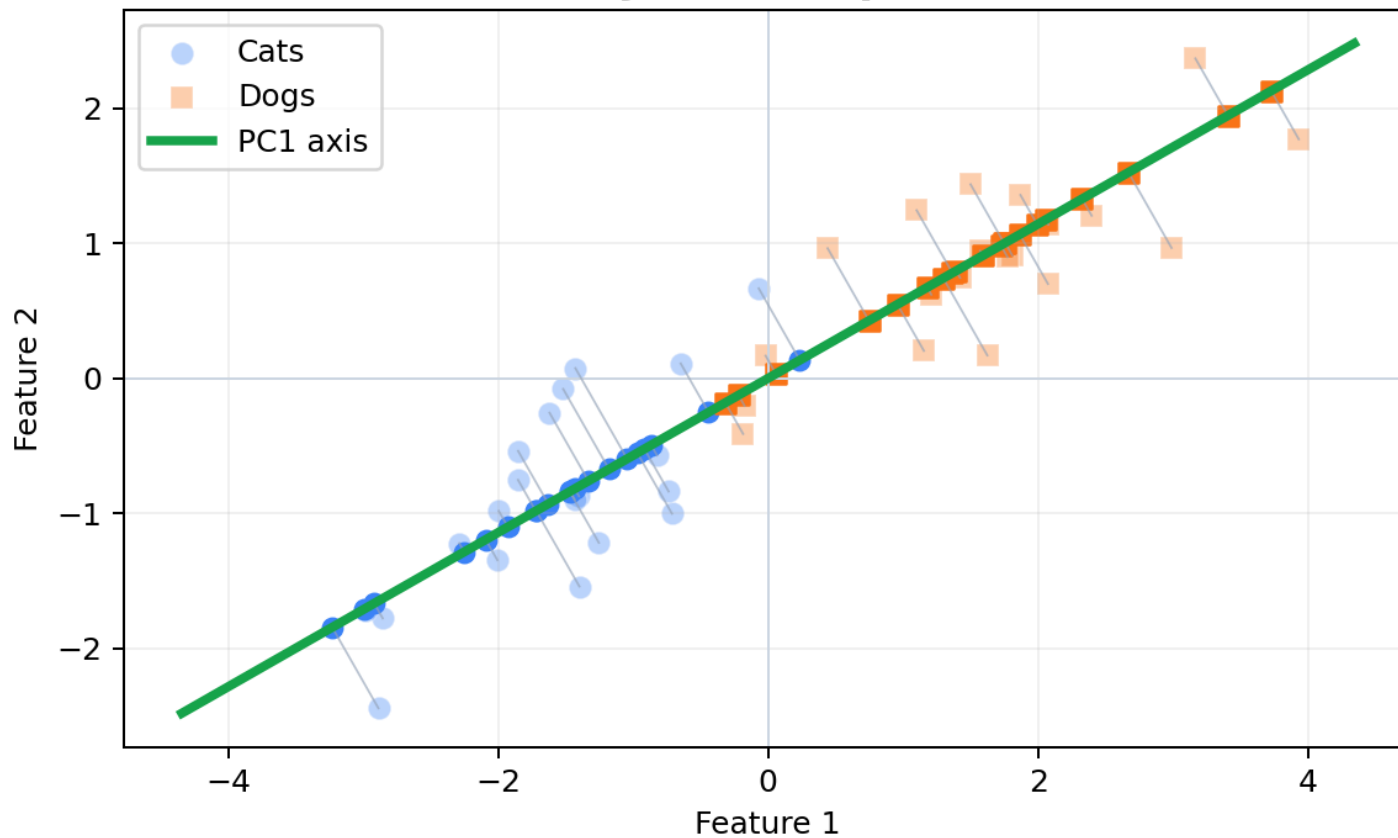
What this state means

- PC1 is the direction of maximum variance.
- PC2 is perpendicular and captures the remaining variance.
- Eigenvector = direction; eigenvalue = amount of variance on that direction.

4. Projection onto PC1

Same points - transformed state by state

State 4 - Project each point onto PC1

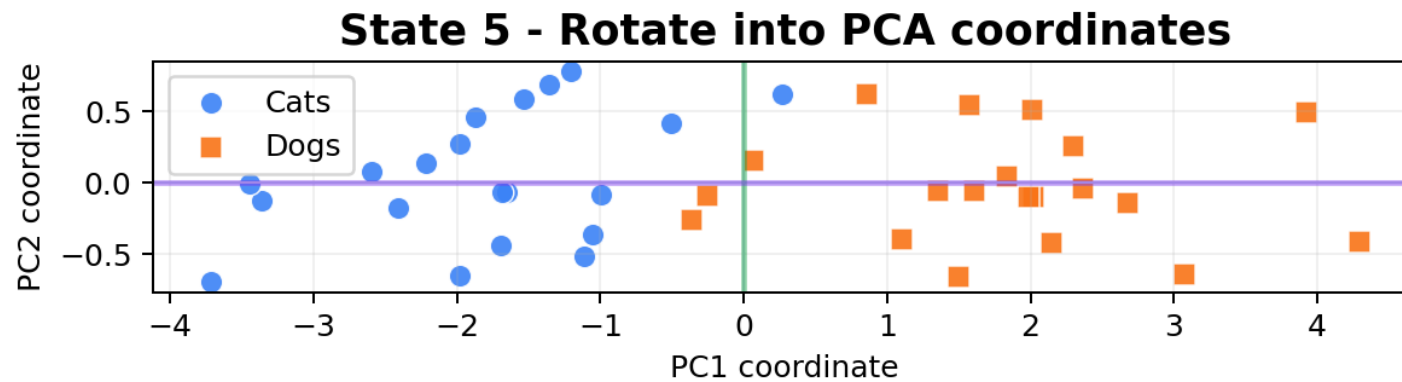


What this state means

- Each point gets a scalar coordinate along PC1.
- Dropping PC2 would compress every point onto the green line.
- Useful structure can survive if most variance is on PC1.

5. Rotate into PCA coordinates

Same points - transformed state by state



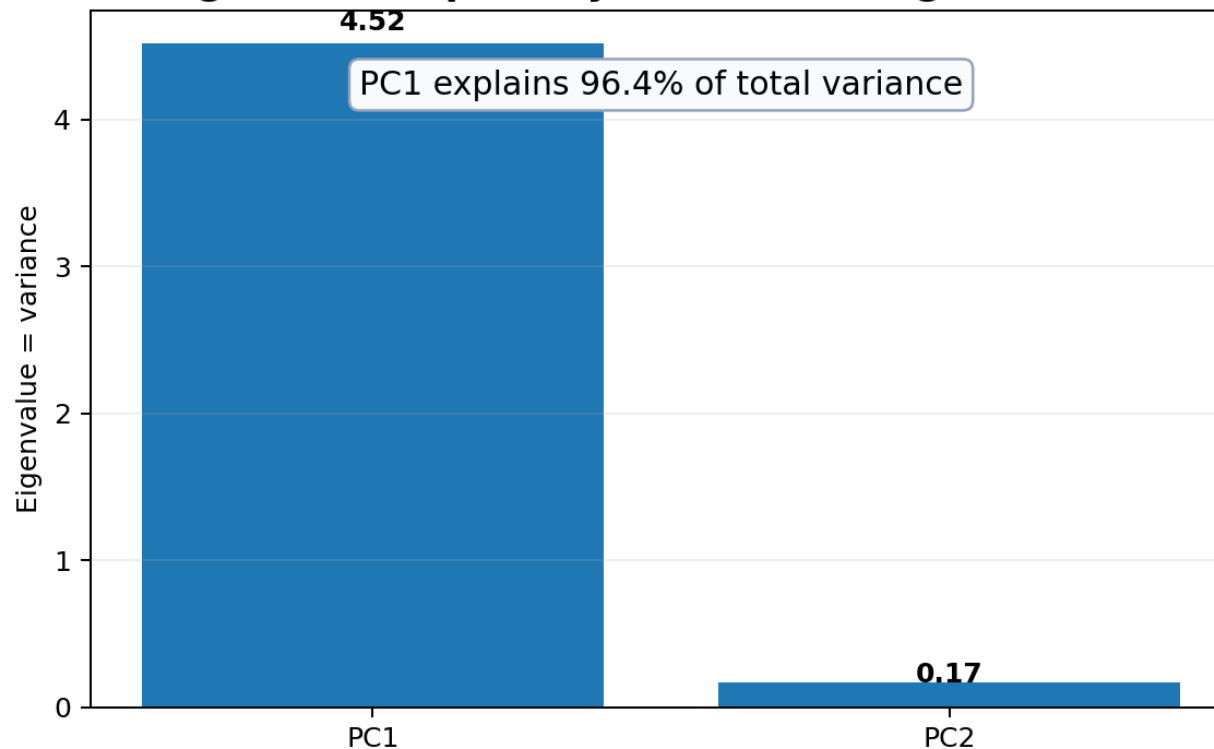
What this state means

- No information is lost when both PCs are retained.
- The axes are now PC1 and PC2 instead of the original features.
- The covariance matrix is diagonal in this basis.

6. Eigenvalues

Same points - transformed state by state

State 6 - Eigenvalues quantify variance along each PCA direction



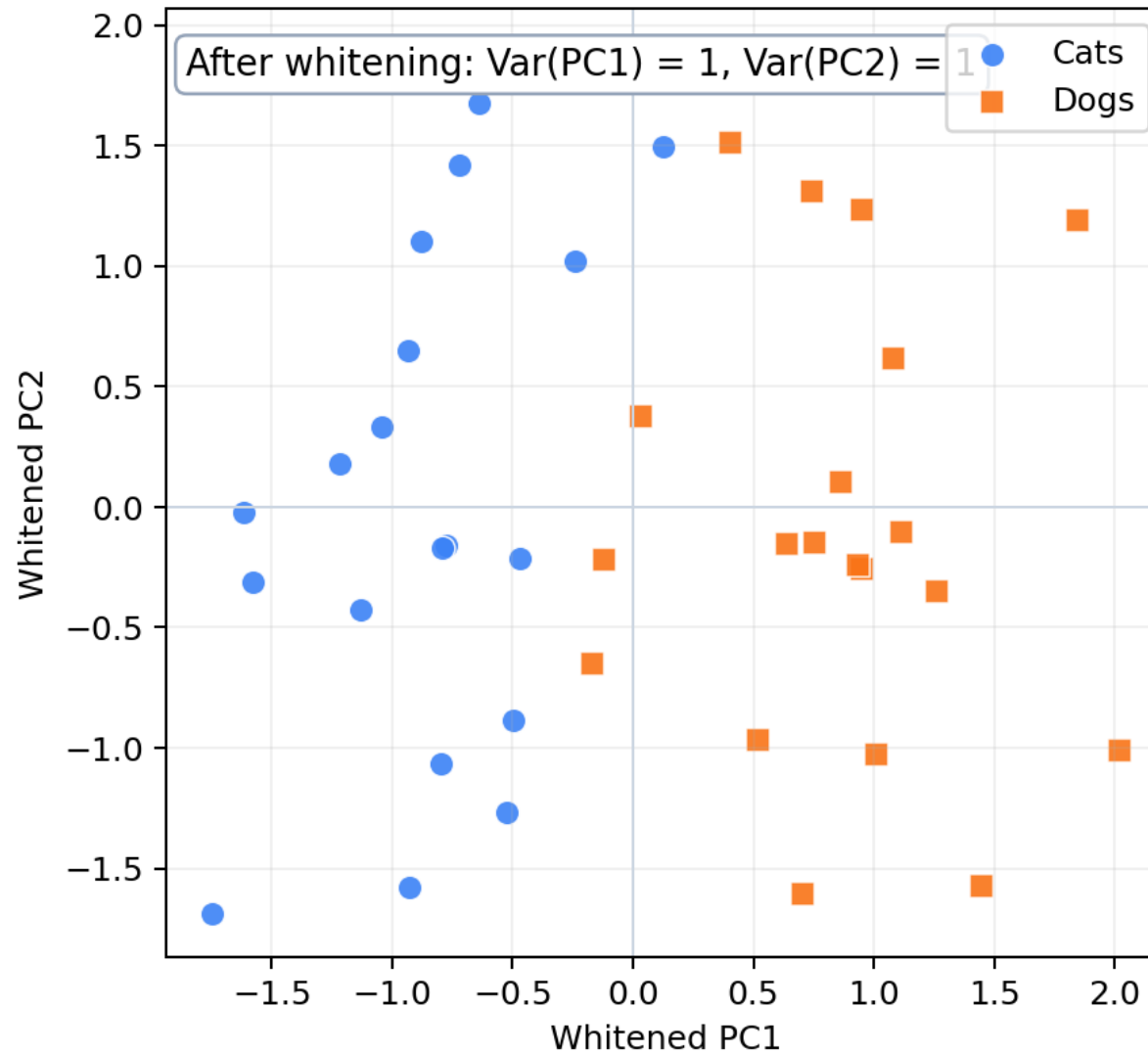
What this state means

- PC1 eigenvalue = 4.52.
- PC2 eigenvalue = 0.17.
- PC1 carries 96.4% of total variance in this example.

7. Whitening

Same points - transformed state by state

State 7 - Whitening rescales each PCA axis



What this state means

- Divide each PCA coordinate by $\sqrt{\text{eigenvalue}}$.
- Large-variance directions shrink more; small-variance directions expand more.
- After whitening, every retained PCA dimension has variance approximately 1.
- Whitening does NOT make cats and dogs identical.

8. The mathematics behind the graphs

Compact reference

Pipeline

Center

$$X_c = X - \text{mean}(X)$$

Covariance

$$C = (1 / (n - 1)) X_c^T X_c$$

Eigendecomposition

$$C v_i = \lambda_i v_i$$

PCA coordinates

$$Z = X_c V$$

Whitening

$$Z_{\text{white}}[:, i] = Z[:, i] / \sqrt{\lambda_i}$$

Key interpretation: eigenvectors tell you WHERE the data varies; eigenvalues tell you HOW MUCH it varies there.

9. Full visual summary

Original -> centered -> PCA -> whitened

The same cat/dog items through the full PCA pipeline

